

NLP Exam Scope

Based on Jurafsky and Martin, *Speech and Language Processing*, 3rd ed. draft
Chapters 2–8 and Appendix B

Prepared for a NLP course

Source and assessment design

You may expect 5 levels of difficulty: Level 1 checks definitions. Level 2 checks recognition of algorithms or modelling procedures. Level 3 checks understanding of algorithms and methods. Level 4 checks interpretation and application. Level 5 checks numerical calculation.

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Chapter 2: Words and Tokens

Exam-topic scope. Students should be able to explain and apply the following topics:

- the role of tokenization as an initial NLP preprocessing step;
- distinctions among word instances, word types, vocabulary size, tokens, characters, morphemes, roots, affixes, inflectional morphology, and derivational morphology;
- the ambiguity of the word concept across modalities and languages, including punctuation, disfluencies, contractions, compounds, languages without whitespace-delimited words, and morphologically rich languages;
- Heaps' Law for vocabulary growth, $|V| = kN^\beta$, and its implications for unknown words and open vocabulary modelling;
- Unicode, code points, encodings, and UTF-8 as foundations for multilingual text processing;
- subword tokenization and the Byte-Pair Encoding (BPE) training and segmentation procedure;
- regular expressions as a formal tool for matching and manipulating text strings;
- deterministic tokenization, sentence segmentation, and common preprocessing choices;
- minimum edit distance, dynamic programming for insertions, deletions, and substitutions, and its relation to word error rate and approximate string matching.

Chapter 3: N-gram Language Models

Exam-topic scope. Students should be able to explain and apply the following topics:

- the definition of a language model as a probability distribution over words, next words, or full sequences;
- the chain rule for sequence probability and the Markov approximation used by n-gram models;
- unigram, bigram, trigram, and general n-gram models;
- maximum-likelihood estimation from corpus counts, including start and end symbols;
- sentence probability under an n-gram model;
- train, development, and test sets, and the danger of evaluating on training data;
- intrinsic evaluation by perplexity and the relationship between probability, cross-entropy, and perplexity;
- text generation by sampling from a language model;
- sparsity and zero-probability problems;
- smoothing and discounting, including add-one smoothing, add- k smoothing, interpolation, and backoff ideas.

Chapter 4: Logistic Regression and Text Classification

Exam-topic scope. Students should be able to explain and apply the following topics:

- supervised text classification and examples such as sentiment analysis, spam detection, language identification, and authorship attribution;
- labels, classes, training instances, feature vectors, weights, and bias terms;

- the binary logistic regression model, weighted sums, sigmoid function, log odds, and decision boundaries;
- feature engineering for text, including bag-of-words, lexicon features, metadata, normalization, and scaling;
- cross-entropy loss and the goal of maximizing conditional likelihood;
- gradient descent and stochastic gradient descent as optimization procedures;
- multinomial logistic regression and the softmax function for multi-class classification;
- train, development, and test splits, cross-validation, and model selection;
- classification evaluation: accuracy, precision, recall, F1, confusion matrices, macro-averaging, and micro-averaging;
- error analysis, feature interpretation, significance testing, and fairness or harm-related considerations in text classification.

Chapter 5: Embeddings

Exam-topic scope. Students should be able to explain and apply the following topics:

- lexical semantics: wordforms, lemmas, senses, synonymy, similarity, relatedness, polysemy, and connotation;
- the distributional hypothesis: words occurring in similar contexts tend to have similar meanings;
- vector semantics and representation learning;
- sparse count-based word representations and dense embedding vectors;
- cosine similarity and nearest-neighbour search in vector space;
- word2vec-style prediction-based embeddings, including skip-gram intuition and negative sampling intuition;
- the distinction between target-word and context-word embeddings;
- subword-informed embeddings such as fastText-style representations;
- embedding visualization, analogy structure, and historical or semantic change;
- intrinsic and extrinsic evaluation of embeddings, and social bias encoded in vector representations.

Chapter 6: Neural Networks

Exam-topic scope. Students should be able to explain and apply the following topics:

- neural units as weighted-input computational units with bias and activation functions;
- common activation functions, including sigmoid, tanh, and ReLU, and their qualitative properties;
- perceptrons, linear separability, and why XOR requires a hidden layer;
- feedforward neural networks and multilayer perceptrons;
- matrix-vector notation for neural-network layers;
- neural classifiers using hand-built text features;

- neural models using embeddings as input representations;
- one-hot vectors and embedding matrices;
- neural language models and their generalization over embeddings;
- computation graphs, forward pass, loss computation, backpropagation, gradients, and parameter updates.

Chapter 7: Large Language Models

Exam-topic scope. Students should be able to explain and apply the following topics:

- large language models as neural models trained to predict tokens from context;
- the distinction among encoder-only, decoder-only, and encoder-decoder architectures at a conceptual level;
- causal or autoregressive generation and next-token prediction;
- prompting, zero-shot prompting, few-shot prompting, and in-context learning as use patterns;
- logits, softmax probabilities, greedy decoding, random sampling, and temperature;
- pretraining as self-supervised learning from text, especially next-token prediction and cross-entropy loss;
- pretraining corpora, scale, data quality, memorization, privacy, and representational skew;
- fine-tuning and task adaptation at a high level;
- evaluation of LLMs by perplexity, downstream task benchmarks, human evaluation, efficiency, and robustness;
- safety and ethics issues, including hallucination, toxicity, bias, unsafe advice, overreliance, and interactional harms.

Chapter 8: Transformers

Exam-topic scope. Students should be able to explain and apply the following topics:

- the transformer as the standard architecture for modern large language models;
- decoder-only causal transformers for left-to-right language modelling;
- token embeddings, positional information, and the residual stream;
- self-attention and the intuition of dynamically selecting relevant context tokens;
- query, key, and value vectors and scaled dot-product attention;
- multi-head attention as parallel attention mechanisms over different representation subspaces;
- causal attention masks and parallel computation over token positions;
- transformer blocks: attention, feedforward layers, residual connections, and layer normalization;
- output projection, logits, softmax, and next-token distributions;
- top- k and top- p sampling, scale, scaling laws, KV caching, parameter-efficient fine-tuning such as LoRA, and interpretability notions such as induction heads and logit lens.

Appendix B: Naive Bayes Classification

Exam-topic scope. Students should be able to explain and apply the following topics:

- text classification tasks such as sentiment analysis, spam detection, language identification, and authorship attribution;
- the naive Bayes model as a probabilistic classifier using Bayes' rule;
- class priors, word likelihoods, posterior probabilities, and the bag-of-words assumption;
- the naive conditional independence assumption and its practical consequences;
- multinomial naive Bayes for document classification;
- training priors and likelihoods from labelled data;
- add-one or add- α smoothing for unseen words;
- classification in log space to avoid numerical underflow and to convert products into sums;
- naive Bayes as a linear classifier under log-probability features;
- evaluation by confusion matrix, accuracy, precision, recall, F1, and macro- or micro-averaging.